

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

MEDICAL DIAGNOSIS

Aim:

To implement a Prolog-based medical diagnosis system using symptoms and logical rules.

Objective:

To infer possible diseases based on symptoms provided by the user.

Algorithm

- 1. Start.**
- 2. Define symptoms.**
- 3. Define disease rules.**
- 4. Enter the patient's symptoms.**
- 5. Match symptoms with disease rules.**
- 6. Display the possible diagnosis.**
- 7. Stop.**

Flowchart

START

|

v

Enter Symptoms

|

v

Match Disease Rules

|

v

Required Symptoms Present?

/ \

Yes No

| |

Disease Check

Derived Next Rule

|

v

Display Diagnosis

|

v

STOP

Prolog Program

% Medical Diagnosis System symptom(ravi, fever).

symptom(ravi, cough). symptom(ravi, body_pain).

**symptom(priya, fever). symptom(priya,
rash).**

**symptom(arun, cough). symptom(arun,
breathlessness).**

**diagnosis(Patient, flu)
:symptom(Patient, fever),
symptom(Patient, cough).**

**diagnosis(Patient, measles)
:symptom(Patient, fever),
symptom(Patient, rash).**

**diagnosis(Patient, pneumonia)
:symptom(Patient, cough),
symptom(Patient, breathlessness).**

Queries

?- diagnosis(ravi, Disease).

Output:

Disease = flu.

?- diagnosis(priya, Disease).

Output:

Disease = measles.

?- diagnosis(arun, Disease).

Output:

Disease = flu.

Result:

Thus, the medical diagnosis system successfully identifies possible diseases based on symptoms.